

Question	Marking Guidance	Mark	Comments
09.1	Pressure increase at source generated by: 1. <u>Sucrose</u> co-tranported into phloem OR <u>Sucrose</u> actively transported into phloem; 2. Causes decrease in water potential (inside phloem); 3. Water enters by osmosis (into phloem);	3 (2 x AO1, 1 x AO2)	1. Ignore facilitated diffusion

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09.2	1. Carbohydrates made by photosynthesis (in the leaf); 2. Less sucrose transported into phloem at leaf/source OR Less sucrose transported to sink/potato/root OR Less sugar stored as starch;	2 (2 x AO2)	1. Accept 'sugars' or any named carbohydrate

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09.3	1. Increased respiration; 2. To provide more ATP for muscle contraction; 3. (More glucose being used so) concentration gradient for glucose OR Increased numbers of glucose transporter proteins in the membrane; 4. (Glucose enters by) facilitated diffusion;	(4 x AO2)	Marks can be awarded across the whole answer 3. For 'transporter proteins' accept channel or carrier proteins

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09.4	Answer key: 4 – The secondary structure is held by hydrogen bonds. The tertiary structure allows the protein to be positioned spanning the whole cell membrane.	1 (AO2)	